

RAHUL SHETTY

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EDUCATION

- University of Colorado, Boulder** - Master of Science, Computer Science | GPA: 3.84/4.0 Aug 2024 – May 2026
- **Coursework:** Machine Learning, Natural Language Processing, Datacenter Scale Computing, Distributed Systems
- Sardar Patel Institute of Technology** - Bachelor of Technology, ECE | GPA: 9.1/10 Aug 2018 – May 2022
- **Coursework:** Data Structures and Algorithms, Programming (Python, Java), Machine Learning

SKILLS

Languages: Python, Java, SQL, BASH scripting, JavaScript, TypeScript, HTML5, CSS3

ML & AI: TensorFlow, PyTorch, Pandas, Scikit-learn, Matplotlib, GenAI, CNN, Transformers, RAG, Knowledge Graphs, VLA

Frameworks & Libraries: React JS, ROS2, NumPy, LangChain, Hadoop, Spark, Streamlit

Technologies: Ansible, Docker, GitHub, Google Cloud Platform, Linux, AWS, ChromaDB, Pinecone, Kubernetes, Firebase

EXPERIENCE

- Graduate Researcher, Human Interaction and Robotics Group – CU Boulder** Jan 2025 – May 2026
- Developed **DEFT, a diffusion-based AI framework** for robot resilience that improved task success from **42% → 99%** and constraint satisfaction from **37% → 75%**, evaluated across **4.7M+ trajectories** and **2,400+ failure scenarios**.
 - Engineered **task-conditioned diffusion models (using FiLM)** to enable zero-shot adaptation to **unseen failures** without retraining, achieving **74% success on out-of-distribution errors** and demonstrating robust autonomy in **real-world hardware tests**.
 - Manuscript accepted at **IEEE International Conference on Robotics and Automation (ICRA) (arXiv:2602.02895)**.
- Software Development Engineer, Nomura** July 2022 – July 2024
- Enhanced the Front Office Supervision (FOS) portal by creating an **organizational hierarchy tree** from data fetched from the **Oracle** database and refactored parts of the **SpringBoot** code.
 - Collaborated with a team to migrate **700+ data marts (13TB)** from **Sybase** to **Snowflake on AWS**, saving over **\$800K in infrastructure costs** while modernizing **three large-scale applications**.
 - Contributed to the development of a **Snowpark framework** using **Python, Shell scripting and SnowSQL** to streamline **Informatica** workflows and cut data processing time by **75%**.
- Software Engineering Intern, Nomura** Jan 2022 – June 2022
- **Automated** retrieval of **Git** details for production releases through **JIRA API integration**, streamlining release preparation and **eliminating human errors**. Developed a scalable **Python** backend with a **Django** web application frontend, supporting **10+ teams**.
 - **Led the migration** of an existing JSP-based monitoring portal to a **React.js** frontend and a refactored **SpringBoot** backend, resulting in a **100% code overhaul**.

PROJECTS

- DriftOps |  | Python, XGBoost, FastAPI, React, TypeScript, MLflow, Docker** July 2026 - Aug 2026
- Built an **ML pipeline** predicting flight delays on **592K+ real BTS flight records**, engineering leakage-safe aggregate features (traffic congestion, weather-delay rate, carrier reliability) via smoothed group statistics, with every run tracked in **MLflow's registry**.
 - Built a **FastAPI** backend (5 REST endpoints) serving **XGBoost** predictions with SHAP-based feature attribution, paired with a **React + TypeScript dashboard** for prediction confidence, model version history, and live drift scores.
 - Built a **retraining pipeline** that auto-promotes candidate models only when **ROC-AUC** beats production, alongside a **self-implemented PSI drift monitor** across all features, supplemented by an **Evidently AI** report.
- Agentic Hybrid RAG Engine |  | Python, Neo4j, Qdrant, LangChain, Docker, Streamlit** Dec 2025 - Jan 2026
- Built a **production-grade Hybrid Retrieval Engine** that routes queries between **Structured (Graph)** and **Unstructured (Vector)** stores, **optimizing latency and relevance** for enterprise use cases.
 - Implemented **Agentic Memory** to personalize responses based on **user roles (CTO vs. CEO)**, adapting technical depth and **boosting relevance scores** by **25%** in **A/B testing** scenarios.
 - Benchmarked system reliability against **4 industry baselines (including BM25 & HyDE)**, demonstrating **3x higher recall** on exhaustive list generation and **100% precision** on fact-verification tasks.
- Photo Memory Finder |  | Python, GCP (Vertex AI, Cloud Run), React, Pinecone** Nov 2025 – Dec 2025
- Architected a **scalable distributed** event-driven pipeline on **Google Cloud Platform** capable of processing concurrent photo uploads with **<2s latency**, achieving semantic understanding beyond simple metadata.
 - Built a **multimodal** ingestion pipeline using **Cloud Pub/Sub** and **Cloud Run** workers to asynchronously process uploads, reducing metadata extraction time by **40%** via parallelized **hybrid embedding generation (image + text)** with **Vertex AI**.
 - Implemented an intelligent LLM Reranking system using **Gemini 2.5 Flash**, filtering and re-ordering vector search results based on query context, improving **retrieval precision by 35%** for complex queries.